AC 209b Milestone 2

Intro and Data Demo

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Project Motivation

- **Core question:** can emotion-related language help explain self-reported MBTI patterns in Reddit writing?
- **Why this project is interesting:**
  - emotion is interpretable, so it gives us more than a black-box text classifier
  - personality labels are noisy, so data quality matters before we commit to modeling
  - the milestone goal is to verify what the data can realistically support
- **Our setup:** use one clean source dataset for emotion supervision and one large target dataset of MBTI-labeled Reddit posts

Data Acquisition and Preparation

- **Emotion dataset (Hugging Face):**
  - 20,000 short texts, 6 balanced emotion labels, 2.35 MB
- **Reddit MBTI dataset (Kaggle):**
  - 13,028,455 posts from 11,773 authors, 16 self-reported MBTI types, 2.70 GB
- **Wrangling steps in the notebook:**
  - renamed columns and standardized MBTI labels
  - removed 180 blank Reddit posts
  - derived the 4 binary dimensions: `E/I`, `N/S`, `F/T`, `J/P`
  - used a fixed 250,000-post sample for computationally expensive text-length EDA
- **Why this matters:** it makes the schema consistent and reveals early that binary dimensions are more realistic than 16-way type prediction

EDA: Clean Source, Difficult Target

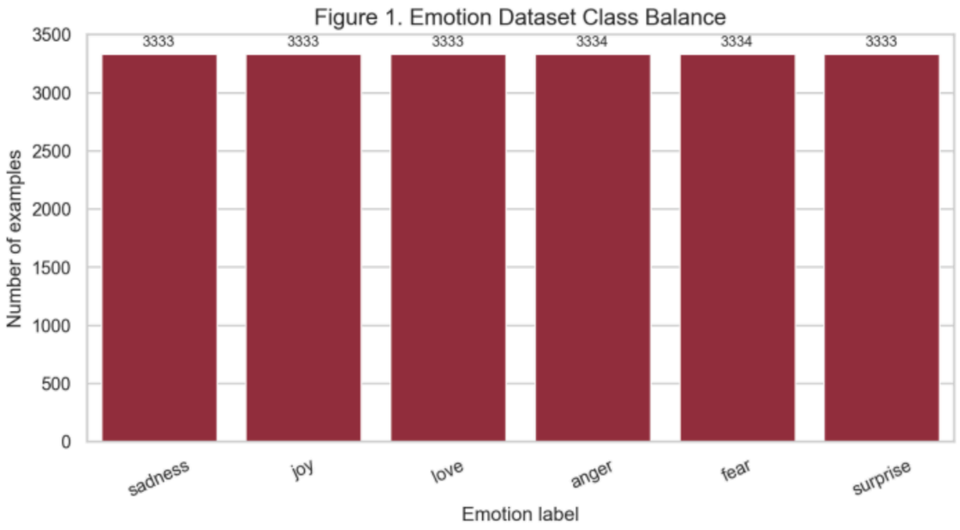

Figure 1: Emotion dataset class balance

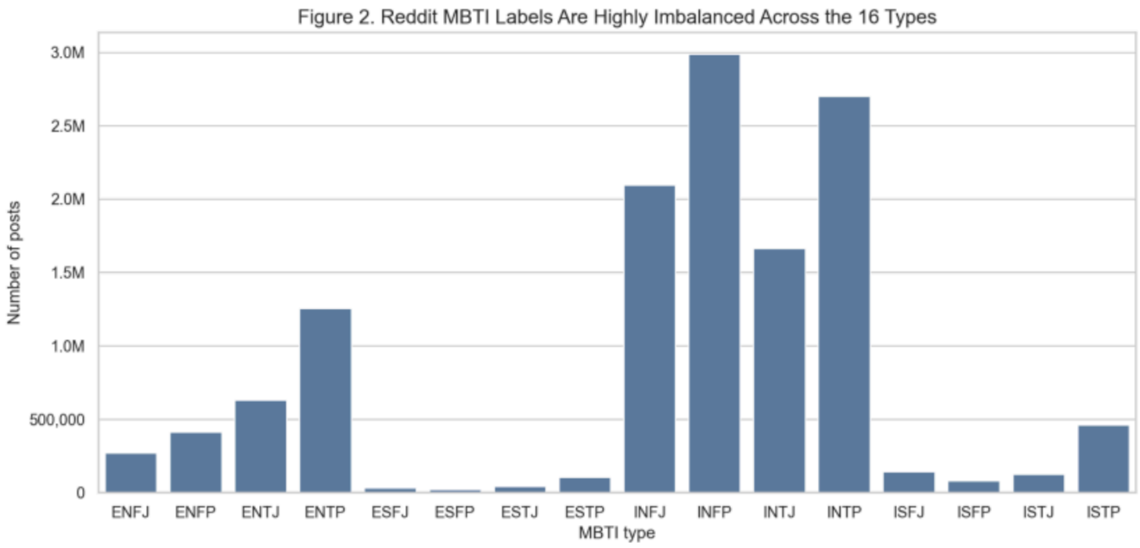

Figure 2: MBTI type imbalance

- The emotion source dataset is almost perfectly balanced, with about 3,333 examples per class.
- The MBTI target dataset is heavily skewed: INFP is **22.94%** of all posts, while ESFP is only **0.18%**.
- **Insight:** the source task is clean, but the target task is much harder.

EDA: Author Concentration Changes the Modeling Plan

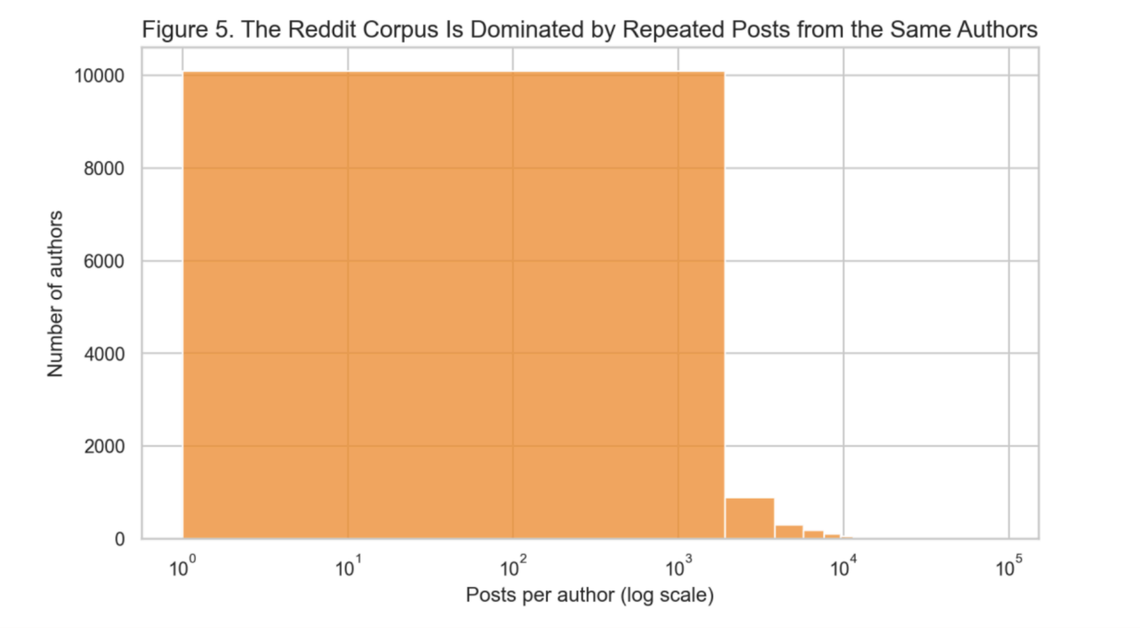

Figure 3: Posts per author in the Reddit MBTI dataset

- The median author contributes **272** posts; the 99th percentile contributes **12,753**.
- The top 1% of authors generate **18.98%** of all posts.
- **Main implication:** train/test splits must be done at the **author level**, not the post level.
- **Handoff:** these EDA results motivate rescoping from 16-way MBTI prediction toward the 4 binary dimensions.

Redefinition and Rescoping

- **Original plan:** 16-type MBTI classification from Reddit text
- **EDA findings:**
  - extreme class imbalance
  - noisy self-reported labels
  - strong author-level clustering
- **Rescope:**
  - move to **4 binary dimensions** (`E/I`, `N/S`, `F/T`, `J/P`)
  - shift from post-level → **user-level analysis**
- **New question:**
  → can emotion profiles explain MBTI dimensions?

Next Steps

- **Emotion model**
  - fine-tune transformer on emotion dataset
  - generate emotion scores for Reddit posts
- **Feature construction**
  - aggregate to **user-level emotion profiles**
- **MBTI analysis**
  - predict each binary dimension
  - compare baseline vs emotion-based features
- **Shared**
  - evaluation, interpretation, final slides

Future Considerations

- **Models**
  - transformer (emotion)
  - logistic regression / random forest (downstream)
- **Challenges**
  - noisy MBTI labels
  - domain shift (Twitter → Reddit)
  - class imbalance + unequal user activity
- **Open questions**
  - is binary rescope sufficient?
  - how to balance interpretability vs performance?